

CENTRIFUGAL PUMP — OEM OPERATION & MAINTENANCE MANUAL

Equipment Tag: Pump-P3 | Model: FlowMaster FP-200 | Rev. 3 (2023)

FlowDynamics Systems Inc. | Document No: FDS-OM-FP200-P3 | CONTROLLED COPY

SECTION 1 — EQUIPMENT OVERVIEW

This manual provides complete operating and maintenance instructions for Pump-P3, a horizontal end-suction centrifugal pump serving the cooling water circulation circuit in Plant Area C. Pump-P3 handles process cooling water at elevated temperatures and must be maintained in accordance with this document to ensure safe and reliable operation.

Equipment Tag: Pump-P3 | Type: Horizontal end-suction centrifugal pump | Model: FlowMaster FP-200 | Serial No: FDS-2020-P3-0089 | Manufacturer: FlowDynamics Systems Inc. | Commissioned: 2020-09-01

SECTION 2 — TECHNICAL SPECIFICATIONS

Rated Flow: 450 GPM at BEP

Rated Head: 120 ft (52 psi)

Maximum Operating Pressure (casing): 200 psi

Design Temperature: 250 °F maximum

Motor: 30 kW, 1,780 RPM, TEFC 480V 3-phase

Impeller Type: Closed, 316 SS | Diameter: 11.5 inches

Seal Type: Single mechanical seal (Flowserve Type 1 equivalent)

Bearing Lubrication: Grease-packed anti-friction bearings (Mobil Polyrex EM)

NPSHr at rated flow: 12 ft

Coupling: Flexible disc type, spacer design for back-pull-out

SECTION 3 — PREVENTIVE MAINTENANCE

3.1 Monthly Checks

- Check bearing temperature (normal: < 180 °F; alarm: 210 °F; trip: 230 °F).
- Inspect mechanical seal for leakage (allowable: < 5 drops/minute).
- Check vibration (normal: < 2.5 mm/s; investigate: > 4.5 mm/s RMS).
- Grease bearings: 15 grams Mobil Polyrex EM per bearing (once per 3 months).

3.2 Annual Overhaul

- Replace mechanical seal assembly regardless of condition.
- Inspect and measure impeller wear ring clearances; replace if > 0.030 in.
- Replace bearing housing seals and repack bearings.
- Perform hydraulic performance test and compare to original curve.

SECTION 4 — EMERGENCY PROCEDURES

4.1 Emergency Shutdown Triggers

- Bearing temperature exceeds 230 °F (auto-trip via TT-P3-001).
- Vibration exceeds 7.5 mm/s RMS (auto-trip via VT-P3-001).
- Mechanical seal failure (visible fluid loss > 50 ml/min).
- Suction pressure drops below 5 psi (cavitation risk — manual ESD).
- Any abnormal noise, smoke, or burning smell.

4.2 Emergency Shutdown Procedure and Post-Trip Access

[WARNING] CRITICAL SAFETY REQUIREMENT — Following emergency shutdown of Pump-P3, a mandatory cooling period of 15 minutes minimum must elapse before any maintenance personnel access the pump casing, mechanical seal area, or bearing housing. This is required because Pump-P3 handles process water at up to 250 °F; residual heat in the casing and trapped pressurised fluid create burn and pressure-release hazards for up to 15 minutes after shutdown. This 15-minute cooling period is NON-NEGOTIABLE and may not be reduced by any field supervisor without written authorisation from the Plant Safety Manager. Failure to comply is a SERIOUS SAFETY VIOLATION.

1. Press red ESD pushbutton P3-ESD-001 (or de-energise at MCC Panel C-08).
2. Close suction isolation valve P3-V-001 immediately after trip.
3. Close discharge check valve P3-V-002 if not auto-actuated.
4. Post a physical barrier and DANGER tag on the pump for the 15-minute cooling period.
5. After 15 minutes: verify casing temperature < 120 °F using infrared thermometer.
6. Depressurise casing via vent/drain before removing any bolted connections.
7. Document ESD event, time, cause, and post-trip inspection in Work Order System.